SSD_UART usage reference

1 Overview

On the SSD20X platform, 4 groups of uarts (including a group of fuarts) are provided, and the pins corresponding to each MODE are as follows:

UART MODE	RX	TX	CTS	RTS
UART0	PM_UART_RX	PM_UART_TX	NA	NA
UART1	PAD_UART1_RX	PAD_UART1_TX	NA	NA
UART2	PAD_GPIO8	PAD_GPIO9	NA	NA
FUART (without flow control)	PAD_FUART_RX	PAD_FUART_TX	NA	NA
FUART (with flow control)	PAD_FUART_RX	PAD_FUART_TX	PAD_FUART_CTS	PAD_FUART_RTS

2. UART platform configuration instructions

2.1. Configuration under Kernel

The main thing that needs to be configured in the uart part is padmux.

2.2. Padmode configuration

1. Add padmux settings, open `kernel\arch\arm\boot\dts\infinity2m-ssc010a-s01a-padmux-display.dtsi`

   Fuart with flow control: use MODE1: the default baud rate is 9600

   ```
   <PAD_FUART_RX          PINMUX_FOR_FUART_MODE_1
   MDRV_PUSE_FUART_RX>,
   <PAD_FUART_TX          PINMUX_FOR_FUART_MODE_1
   MDRV_PUSE_FUART_TX>,
   <PAD_FUART_CTS        PINMUX_FOR_FUART_MODE_1
   MDRV_PUSE_FUART_CTS>,
   <PAD_FUART_RTS        PINMUX_FOR_FUART_MODE_1
   MDRV_PUSE_FUART_RTS>,
   ```

   Fuart不带流控：使用MODE2：默认波特率是9600 // /dev/ttyS2

   ```
   <PAD_FUART_RX     PINMUX_FOR_FUART_MODE_2
   MDRV_PUSE_FUART_RX>,
   <PAD_FUART_TX     PINMUX_FOR_FUART_MODE_2
   MDRV_PUSE_FUART_TX>,
   ```

   uart0配置：默认波特率是115200 // /dev/ttyS0

   ```
   <PM_UART_RX    PINMUX_FOR_UART0_MODE_1    MDRV_PUSE_UART0_RX >,
   <PM_UART_TX    PINMUX_FOR_UART0_MODE_1    MDRV_PUSE_UART0_TX >,
   ```

   uart1配置：默认波特率是9600 // /dev/ttyS1

```
<PAD_UART1_RX     PINMUX_FOR_UART1_MODE_1    MDRV_PUSE_UART1_RX>,
<PAD_UART1_TX     PINMUX_FOR_UART1_MODE_1    MDRV_PUSE_UART1_TX>,
```

uart2配置：默认波特率是9600 // /dev/ttyS3

```
<PAD_GPIO8     PINMUX_FOR_UART2_MODE_2     MDRV_PUSE_UART2_RX>,
<PAD_GPIO9     PINMUX_FOR_UART2_MODE_2     MDRV_PUSE_UART2_TX>,
```

配置完成后，重新编译kernel，替换kernel image到project烧录。

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3. UART使用说明

3.1. 确认uart pin使用的/dev/ttySx

通过查看dts，确认uart对应的是哪路serialx，serialx对应的就是/dev/ttySx

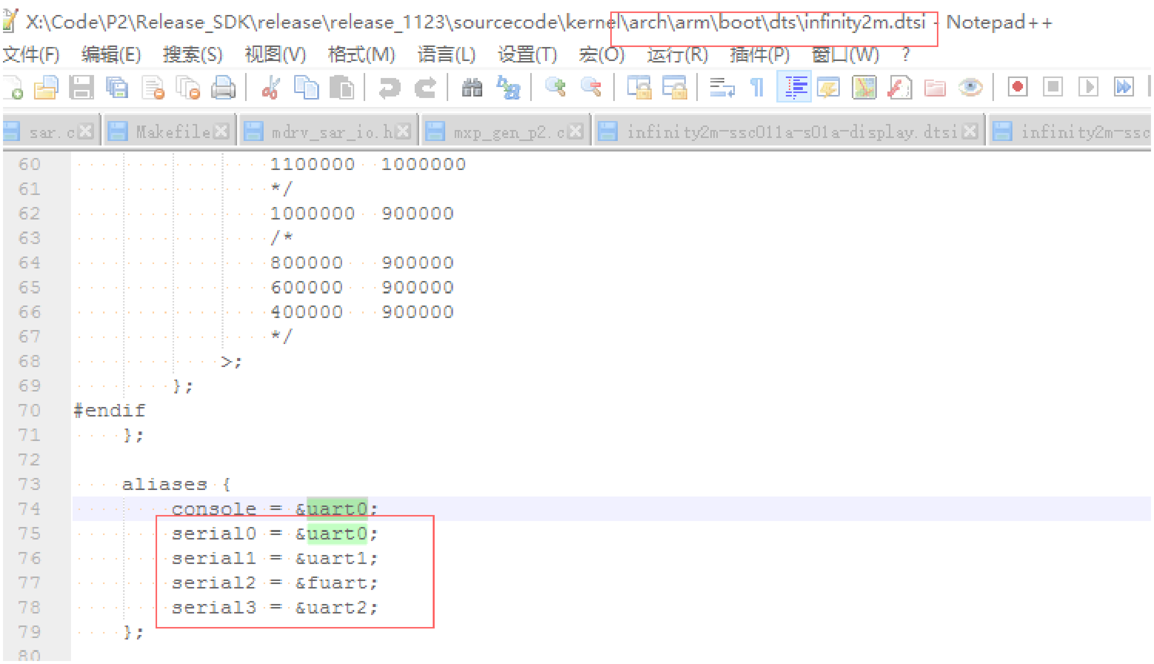

图3-1 查看serialx

3.2. 应用层如何使用

应用层使用参考如下链接：

https://blog.csdn.net/aaron_111/article/details/84876508
[https://blog.csdn.net/aaron_111/article/details/84876508]

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4. uart驱动架构及debug方法

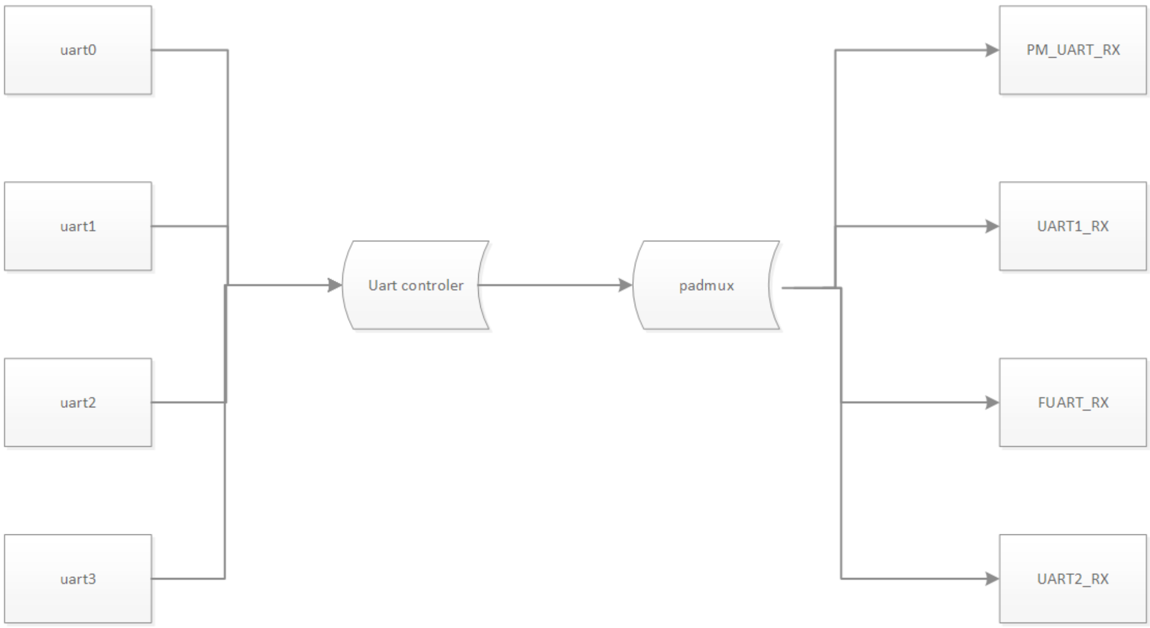

图4-1 Uart驱动主要框架图

从上图可以看出来驱动的主要工作就是把uart controller和对应的pad串接起来。

可以通过以下寄存器确定对应的uart controller和padmux有没有串起来：

- Uart controller

  0x101e 0x53(16bit)

h0053	h0053	3	0	reg_uart_sel0	3	0	4	h0	rw	Select controller for PAD_PM_UART_RX and PAD_PM_UART_TX 0000: N/A 0001: FUART 0010: UART0 0011: UART1 0100: UART2 0101: UART_DEC Note: For PAD_PM_UART_RX and PAD_PM_UART_TX, please refer to the "reg_hk51_uart0_en" and "reg_uart_rx_enable" in reg_pm_sleep.xls (a). "reg_hk51_uart0_en" == 0 (b). "reg_uart_rx_enable" == 1
h0053	h0053	7	4	reg_uart_sel1	3	0	4	h1	rw	Select controller for PAD_FUART_RX and PAD_FUART_TX

图4-2

h0053	h0053	15	12	reg_uart_sel3	3	0	4	h3	rw	Select controller for PAD_UART1_RX and PAD_UART1_TX
h0054	h0054	3	0	reg_uart_sel4	3	0	4	h4	rw	Select controller for PAD_UART2_RX and PAD_UART2_TX

图4-3

- Padmux

0x101e 0x03(16bit)

h0003	h0003	2	0	reg_fuart_mode	2	0	3	h0	rw	FUART Mode
h0003	h0003	6	4	reg_uart0_mode	2	0	3	h0	rw	UART0 Mode
h0003	h0003	10	8	reg_uart1_mode	2	0	3	h0	rw	UART1 Mode
h0003	h0003	14	12	reg_uart2_mode	2	0	3	h0	rw	UART2 Mode

图4-4